

The switching conjecture for main eigenvalues holds for trees of diameter at most 4^*

Horace Dong

Independent researcher

`maxwelldhx@gmail.com`

ORCID: [0009-0002-7554-7548](https://orcid.org/0009-0002-7554-7548)

September 2026 (version 2)

Abstract

An eigenvalue of a signed graph is *main* if its eigenspace is not orthogonal to the all-ones vector. Akbari, França, Ghasemian, Javarsineh and de Lima conjectured, in the form given by Akbari, Kumar, Mohar and Pragada, that every connected graph other than K_2 and $K_4 - e$ has a switching, a choice of a sign ± 1 at each vertex, for which all eigenvalues of the resulting signed graph are main, and Akbari, Kumar, Mohar and Pragada asked for a proof for trees and for connected regular graphs. We prove the conjecture for every tree of diameter at most 4 other than K_2 ; this class contains the stars, the double stars and the harmonic trees, for which it was known. Apart from a few explicit ones, the eigenvalues of such a tree are the numbers $\pm\sqrt{t}$ for the roots t of a secular equation. If all neighbours of the centre receive the same sign, each of these whose minimal polynomial is even is main; along two explicit families of switchings, each remaining pair of Galois orbits $\pm O$ spoils at most four members, at most two when its eigenvalues are irrational, and an even orbit spoils at most one. This settles every tree in which a neighbour of the centre has at least 13 leaves; the other cases reduce to explicit switchings and to a finite set of 13,376 multisets of branch sizes, which the Lean kernel checks. Every result of the paper, including the signature version of Theorem 1, is formally verified in Lean 4. The results were found and formalised by AI coding agents working under the author’s direction; their role is described at the end of the paper.

1 Introduction

Background. Let G be a finite simple graph with adjacency matrix $A = A(G)$. A *switching* of G is a vector $\mathbf{s} \in \{\pm 1\}^{V(G)}$. The signed graph $G^{\mathbf{s}}$ has the edges of G , the edge uv carrying the sign $s_u s_v$, so that $A(G^{\mathbf{s}}) = D_{\mathbf{s}} A D_{\mathbf{s}}$, where $D_{\mathbf{s}}$ is the diagonal matrix with diagonal $\mathbf{s}$ [2]. An eigenvalue of a signed graph is *main* if its eigenspace is not orthogonal to the all-ones vector $\mathbf{1}$ [3, 9]. A regular graph has only one main eigenvalue [9], whereas almost all graphs have only main eigenvalues [8]. Akbari, França, Ghasemian, Javarsineh and de Lima [1] asked whether a suitable switching always makes all eigenvalues main, and conjectured that it does. We state the conjecture in the form of [2, Conjecture 1.1].

Conjecture ([1]). *Let G be a connected graph other than K_2 and $K_4 - e$. Then some switching $\mathbf{s}$ of G makes every eigenvalue of $G^{\mathbf{s}}$ main.*

* *Version 2.* Every result of this version is formally verified in Lean 4 (Section 7), including the finite search that version 1 carried out with three programs. Computational claims of version 1 for which we have no formal proof have been deleted: the supporting evidence on trees with at most 24 and 44 vertices, the computed values in Remark 8.1, and some counts in Section 6. A few other sentences have been reworded or removed; the theorems are unchanged. The main changes are listed at the end of Section 7.

Only distinct eigenvalues matter [2, Proposition 1.3]. The abstract of [1] states the conjecture for every graph other than K_2 and $K_4 - e$; Akbari, Kumar, Mohar and Pragada [2] state it for connected graphs and show that connectedness is needed (no switching of $K_2 \cup K_1$ works). For trees the two versions agree.

According to its abstract, [1] proves the conjecture for Cayley, distance-regular, vertex- and edge-transitive graphs, double stars and paths. It also covers complete bipartite graphs, hence stars ([1, Theorem 4.1], as cited in [10, Section 4]), and, by a computer search, all connected graphs on at most nine vertices other than K_2 and $K_4 - e$ (as reported in [2, Section 1.2] and [10, Section 5]). Shao and Yuan [10] proved it for harmonic trees [10, Proposition 2.1], complete multipartite graphs and the graphs $S_{n,r}$, and Stanić [11] gave sufficient conditions for net-regular signed graphs. The authors of [2] proved two asymptotic versions, with $n - O(n/(\log n)^{1/4})$ main eigenvalues counted with multiplicity and with $d - O(d/(\log d)^{1/4})$ distinct main eigenvalues, where n is the order and d the number of distinct eigenvalues. They single out regular graphs and trees as unresolved cases and, in [2, Problem 1.6], ask to show that the conjecture “holds for trees and connected regular graphs of order $n \geq 3$ ”.

Result.

Theorem 1. *Let T be a tree of diameter at most 4 other than K_2 . Then some switching $\mathbf{s}$ of T makes every eigenvalue of $T^{\mathbf{s}}$ main.*

The trees of diameter at most 4 are the trees of radius at most 2. In the notation of Section 3 (the identifications are immediate from the definitions) they include the stars $T(0, \dots, 0)$, the double stars $T(b, 0, \dots, 0)$, the paths $P_3 = T(0, 0)$, $P_4 = T(1, 0)$ and $P_5 = T(1, 1)$, and the harmonic trees of [10], in which a vertex of degree $a^2 - a + 1$ has only neighbours of degree a and all other vertices are leaves, that is, the trees $T(a - 1, \dots, a - 1)$ with $a^2 - a + 1$ branches. So Theorem 1 contains the known cases above of diameter at most 4 and adds all other trees of diameter 4. Since every signature of a tree is a switching of the all-positive one, Theorem 1 also gives the signature version of the conjecture [2, Conjecture 1.7] for these trees: for every tree T of diameter at most 4 other than K_2 and every choice of a sign on each edge of T , some switching makes every eigenvalue of the signed tree main.

Method. A tree of diameter at most 4 with at least three vertices is a rooted tree $T(\mathbf{a})$ of height at most 2: a centre c with neighbours $v_1, \dots, v_k$, where v_i has a_i further neighbours, all of them leaves. Apart from 0 and the numbers $\pm\sqrt{b}$ for branch sizes $b \geq 1$ that occur more than once, its eigenvalues are the numbers $\pm\sqrt{t}$ for the roots t of a *secular polynomial* R (Lemma 3.1). Whether an eigenvalue becomes main after a switching is a simple condition (Lemma 3.2), which for the secular eigenvalues depends only on the Galois orbit. If all neighbours of the centre receive the same sign, every eigenvalue in such an orbit whose minimal polynomial is even is main (Lemma 3.4 and the criterion after Lemma 3.2); the other orbits form *non-even pairs* $\pm O$, with $\theta \in \mathbb{Q}(\theta^2)$ for $\theta \in O$, which are *integer* pairs if $O \subseteq \mathbb{Z}$ and *irrational* pairs otherwise. Along two explicit families of switchings, the *rows* of Section 4, an integer pair spoils at most four members and an irrational pair at most two, by a linear-independence argument; so a row that is long compared with the number of non-even pairs contains a good switching (Corollary 4.4). There are at most $\lfloor \sqrt{b^* - 1} \rfloor + 1$ integer pairs, because they come from perfect squares that are roots of R , and all roots of R but one are smaller than the largest branch size $b^* = \max_i a_i$ (Lemma 5.2). For $b^* \geq 13$ this makes the row of the largest branches long enough (Proposition 5.3). For $2 \leq b^* \leq 12$ the trees not covered by the rows are trees $T(\mathbf{a})$ with $\mathbf{a}$ in an explicit finite set of 13,376 multisets, and a finite search shows that each of them satisfies one of the two criteria of Corollary 4.4, except possibly three trees with at most seven vertices, for which we give switchings (Propositions 5.5 and 5.6). Trees with $b^* \leq 1$ are treated by explicit switchings (Proposition 5.1).

Scope. One step of the proof, Proposition 5.5, is a finite computation. In version 1 it was carried out by three independently written programs (Section 6); in version 2 the Lean kernel checks it, and every result of the paper is formally verified (Section 7). The problem remains open for trees in general and for regular graphs; our argument relies on the explicit spectrum of Lemma 3.1.

Prior work. The full text of [1] was not accessible to us: the publisher’s page refused automated access, and we found no other copy. Our account of its results rests on its abstract, as shown by zbMATH Open, and on the summaries in [10] and [2]. The problem appeared in version 1 of [2], submitted to the arXiv on 22 September 2026. We searched for earlier or concurrent work on the conjecture for trees, in particular for trees of diameter at most 4, on 26 September 2026 between 18:48 and 19:49 UTC and between 21:38 and 22:01 UTC, briefly again at 23:00 UTC, and on 27 September 2026 between 00:37 and 00:42 UTC; the last search was made by the referee agent described in the disclosure. The sources were: the arXiv (the listing of [2], which had only version 1, the arXiv API, and the new submissions in math.CO); zbMATH Open, Semantic Scholar, OpenAlex and Crossref, including the works citing [1] and [10]; the problem database MathDB (<https://mathdb.com>), whose entry for the conjecture was still open, with no solutions; MathOverflow and Mathematics Stack Exchange; GitHub code, commits, issues, pull requests and repositories; public repositories of AI-assisted work on open problems, among them `trureturing` [12]; the publication lists of two authors of [1]; and web search. We did not search Google Scholar or MathSciNet, and dblp and some repository search engines could not be queried, because they returned checks for automated access, which we did not bypass. We found no proof or claim of the conjecture for all trees of diameter at most 4, or for a class of trees containing them; the tree classes treated in the work we found are those listed above. According to its abstract, [5] determines the unsigned trees of diameter 4 with exactly three main eigenvalues; it does not concern switchings. A final check on 27 September 2026 between 02:04 and 02:06 UTC (the arXiv, where [2] still had only version 1; GitHub; MathDB; and MathOverflow) found nothing new. Before the release of version 2 we repeated the searches on 29 September 2026 between 18:35 and 19:35 UTC (the arXiv, where [2] still had only version 1; Semantic Scholar, OpenAlex, Crossref and zbMATH Open, including the works citing [2] and [1]; MathDB, whose entry for the conjecture was still open; MathOverflow and Mathematics Stack Exchange; GitHub, including [12]; Zenodo; and web search) and again found no proof or claim of the conjecture for the trees of diameter at most 4 or for a class of trees containing them. This reports the coverage of our searches, not a guarantee of priority.

2 Switchings and main eigenvalues

Let A be a real symmetric $n \times n$ matrix. For an eigenvalue θ of A let E_θ be its eigenspace and P_θ the orthogonal projection onto E_θ . Let $\mathbf{s} \in \{\pm 1\}^n$.

Lemma 2.1. *An eigenvalue θ of A is a main eigenvalue of $D_{\mathbf{s}}AD_{\mathbf{s}}$ if and only if $P_\theta\mathbf{s} \neq 0$.*

Proof. As $D_{\mathbf{s}}^2 = I$, the matrix $D_{\mathbf{s}}AD_{\mathbf{s}}$ has the same eigenvalues as A , with eigenspaces $D_{\mathbf{s}}E_\theta$ and projections $D_{\mathbf{s}}P_\theta D_{\mathbf{s}}$. The eigenspace $D_{\mathbf{s}}E_\theta$ is orthogonal to $\mathbf{1}$ if and only if the projection $D_{\mathbf{s}}P_\theta D_{\mathbf{s}}\mathbf{1} = D_{\mathbf{s}}P_\theta\mathbf{s}$ of $\mathbf{1}$ onto it vanishes. $\square$

We call $\mathbf{s}$ *good* for A , or for a graph G with $A = A(G)$, if $P_\theta\mathbf{s} \neq 0$ for every eigenvalue θ of A . By Lemma 2.1, Theorem 1 says that every tree of diameter at most 4 other than K_2 has a good switching. If $\mathbf{s}$ is good, so is $-\mathbf{s}$.

Lemma 2.2 (Krylov certificate). *Let A be a symmetric integer matrix with d distinct eigenvalues, let $\mathbf{s} \in \mathbb{Z}^n$, and let K be the integer matrix with columns $\mathbf{s}, A\mathbf{s}, \dots, A^{d-1}\mathbf{s}$. Then $\text{rank}_{\mathbb{Q}} K$ is the number of eigenvalues θ with $P_{\theta}\mathbf{s} \neq 0$. In particular, $\mathbf{s}$ is good if K has rank d over $\mathbb{F}_p$ for some prime p .*

Proof. We have $A^j\mathbf{s} = \sum_{\theta} \theta^j P_{\theta}\mathbf{s}$, the nonzero vectors $P_{\theta}\mathbf{s}$ are orthogonal, and the Vandermonde matrix $(\theta^j)_{\theta, 0 \leq j < d}$ is invertible. So the column space of K is spanned by the vectors $P_{\theta}\mathbf{s}$, and its dimension is the number of nonzero ones; the rank of an integer matrix is the same over $\mathbb{Q}$ and over $\mathbb{R}$. Reduction modulo p does not increase the rank. $\square$

3 The spectrum of a tree of diameter at most 4

Notation. Let $\mathbf{a} = (a_1, \dots, a_k)$, with $k \geq 1$, be a multiset of nonnegative integers. The tree $T(\mathbf{a})$ has a vertex c , the *centre*, adjacent to $v_1, \dots, v_k$, and v_i is adjacent to a_i further vertices, the leaves in the set L_i . So $T(\mathbf{a})$ has $n = 1 + k + \sum_i a_i$ vertices. We call v_i together with L_i the i th *branch*; it is *bare* if $a_i = 0$. Let B be the set of distinct values of the a_i , let $r = |B|$ and $b^* = \max B$, and for $b \in B$ let $I_b = \{i : a_i = b\}$ and $k_b = |I_b|$; we put $k_0 = 0$ if $0 \notin B$. We write $B = \{b_1 < \dots < b_r\}$.

A tree with $n \geq 3$ vertices and diameter at most 4 has a vertex of eccentricity at most 2; rooting the tree at such a vertex shows that it is some $T(\mathbf{a})$. Conversely, every $T(\mathbf{a})$ has diameter at most 4. The results of this section and the next hold for every multiset $\mathbf{a}$, so the choice of representation does not matter.

Let

$$F(t) = \sum_{b \in B} \frac{k_b}{t - b}, \quad R(t) = \prod_{b \in B} (t - b) - \sum_{b \in B} k_b \prod_{b' \in B \setminus \{b\}} (t - b') = \prod_{b \in B} (t - b) (1 - F(t)).$$

Then $R \in \mathbb{Z}[t]$ is monic of degree r ; we call it the *secular polynomial* of $T(\mathbf{a})$.

Lemma 3.1 (spectrum). *Let $T = T(\mathbf{a})$ and $A = A(T)$.*

- (i) *R has r simple roots $t_1 < \dots < t_r$, with $b_j < t_j < b_{j+1}$ for $1 \leq j < r$ and $t_r > b_r$; in particular $t_j > 0$ and $t_j \notin B$. For each j , both $\theta = \sqrt{t_j}$ and $\theta = -\sqrt{t_j}$ are simple eigenvalues of A , with eigenvector x^{θ} given by $x_c^{\theta} = 1$, $x_{v_i}^{\theta} = \theta/(t_j - a_i)$ and $x_{\ell}^{\theta} = 1/(t_j - a_i)$ for $\ell \in L_i$. We call these $2r$ eigenvalues secular.*
- (ii) *For $b \in B$ with $b \geq 1$ and $k_b \geq 2$, both $\theta = \pm\sqrt{b}$ are eigenvalues of multiplicity $k_b - 1$, and E_{θ} consists of the vectors with $x_{v_i} = \alpha_i$ and $x_{\ell} = \alpha_i/\theta$ for $\ell \in L_i$ and $i \in I_b$, where $\sum_{i \in I_b} \alpha_i = 0$, and with all other coordinates 0.*
- (iii) *If $k_0 \geq 1$, then $\ker A$ consists of the vectors with $x_c = 0$, with $x_{v_i} = 0$ and $\sum_{\ell \in L_i} x_{\ell} = 0$ whenever $a_i \geq 1$, and with $\sum_{i \in I_0} x_{v_i} = 0$. If $k_0 = 0$, then $\ker A$ consists of the vectors with $x_{v_i} = 0$ and $\sum_{\ell \in L_i} x_{\ell} = -x_c$ for every i .*
- (iv) *A has no other eigenvalues. Hence A has $d = 2r + 2|\{b \in B : b \geq 1, k_b \geq 2\}| + [\ker A \neq 0]$ distinct eigenvalues.*

Proof. The equation $Ax = \theta x$ reads

$$\theta x_{\ell} = x_{v_i} \ (\ell \in L_i), \quad \theta x_{v_i} = x_c + \sum_{\ell \in L_i} x_{\ell}, \quad \theta x_c = \sum_{i=1}^k x_{v_i}. \quad (3.1)$$

Let $\theta \neq 0$ and $x \neq 0$ be a solution. The first two equations give $x_{\ell} = x_{v_i}/\theta$ and $(\theta^2 - a_i)x_{v_i} = \theta x_c$. If $x_c \neq 0$ we may take $x_c = 1$; then $\theta^2 \neq a_i$ for all i , $x_{v_i} = \theta/(\theta^2 - a_i)$, and the third equation

becomes $F(\theta^2) = 1$, that is, $R(\theta^2) = 0$. Conversely, these formulas solve (3.1) whenever $R(\theta^2) = 0$ and $\theta^2 \notin B$. If $x_c = 0$, then $x_{v_i} = 0$ unless $a_i = \theta^2$, so $\theta^2 = b \in B$ with $b \geq 1$, and the third equation gives $\sum_{i \in I_b} x_{v_i} = 0$; this is (ii), and it needs $k_b \geq 2$. For $\theta = 0$ the equations say $x_{v_i} = 0$ when $a_i \geq 1$, $x_c + \sum_{\ell \in L_i} x_\ell = 0$ for every i , and $\sum_i x_{v_i} = 0$; a bare branch forces $x_c = 0$, and (iii) follows.

The value $R(b_j) = -k_{b_j} \prod_{i \neq j} (b_j - b_i)$ has sign $(-1)^{r-j+1}$, and $R(t) \rightarrow +\infty$ as $t \rightarrow \infty$. So R changes sign on each interval (b_j, b_{j+1}) and on (b_r, ∞) , which gives r roots as stated; there are no others, since $\deg R = r$. As no t_j lies in B , the eigenvalues in (i), (ii) and (iii) are distinct, and those in (i) are simple. Finally the dimensions add up to n : they are $2r$ in (i), $2 \sum_{b \geq 1} (k_b - 1)$ in (ii), and $\sum_{a_i \geq 1} (a_i - 1) + k_0 - 1$ if $k_0 \geq 1$, or $1 + \sum_i (a_i - 1)$ if $k_0 = 0$, in (iii); in both cases the total is $1 + k + \sum_i a_i$. $\square$

Moreover $t_r \leq b^* + k$: for $t > b^* + k$ all $t - b$ with $b \in B$ are positive, so $F(t) \leq k/(t - b^*) < 1$ and $R(t) > 0$. (Version 1 used this bound, without stating it, for the search range of a program in Section 6.)

The effect of a switching. For a switching $\mathbf{s}$ of $T(\mathbf{a})$ we write $s_c = s(c)$, $\sigma_i = s(v_i)$ and $\lambda_i = \sum_{\ell \in L_i} s(\ell)$, the *leaf sum* of branch i ($\lambda_i = 0$ for a bare branch), and for $b \in B$

$$S_b = \sum_{i \in I_b} \sigma_i, \quad \Lambda_b = \sum_{i \in I_b} \lambda_i, \quad A_b = s_c k_b + \Lambda_b.$$

Lemma 3.2 (main eigenvalues after switching). *A switching $\mathbf{s}$ of $T = T(\mathbf{a})$ is good if and only if the following three conditions hold.*

(S) *For every secular eigenvalue θ , with $t = \theta^2$,*

$$G_{\mathbf{s}}(\theta) := \sum_{b \in B} \frac{A_b + S_b \theta}{t - b} \neq 0.$$

(L) *For every $b \in B$ with $b \geq 1$ and $k_b \geq 2$ and each $\theta \in \{\sqrt{b}, -\sqrt{b}\}$, the numbers $\sigma_i + \lambda_i/\theta$ with $i \in I_b$ are not all equal.*

(Z) *If 0 is an eigenvalue: when $k_0 \geq 1$, some branch with $a_i \geq 2$ has leaves of both signs, or the σ_i with $i \in I_0$ are not all equal; when $k_0 = 0$, some branch has leaves of both signs, or $s_c \neq \sum_i \varepsilon_i$, where ε_i is the common sign of the leaves in L_i .*

Proof. By Lemma 3.1, $\mathbf{s}$ is good if and only if $\mathbf{s}$ is not orthogonal to x^θ for each secular θ , nor to $E_{\pm\sqrt{b}}$, nor to $\ker A$. First, $\mathbf{s} \cdot x^\theta = s_c + \sum_i (\sigma_i \theta + \lambda_i)/(t - a_i)$; writing $s_c = s_c F(t)$ and grouping the branches by size gives $G_{\mathbf{s}}(\theta)$. Second, for $x \in E_\theta$ as in Lemma 3.1(ii) we have $\mathbf{s} \cdot x = \sum_{i \in I_b} \alpha_i (\sigma_i + \lambda_i/\theta)$, which vanishes for all α with $\sum \alpha_i = 0$ exactly when the coefficients are all equal. Third, let $k_0 \geq 1$. By Lemma 3.1(iii), $\mathbf{s} \perp \ker A$ if and only if $\mathbf{s}$ is constant on each L_i with $a_i \geq 2$ and on $\{v_i : i \in I_0\}$. Let $k_0 = 0$. If two leaves ℓ, ℓ' of one branch have opposite signs, then $e_\ell - e_{\ell'} \in \ker A$ and $\mathbf{s} \cdot (e_\ell - e_{\ell'}) \neq 0$. Otherwise $\mathbf{s} \cdot x = x_c (s_c - \sum_i \varepsilon_i)$ for $x \in \ker A$, and $\ker A$ contains a vector with $x_c = 1$. $\square$

The first step of the proof shows more: a secular eigenvalue θ is a main eigenvalue of $D_{\mathbf{s}} A D_{\mathbf{s}}$ if and only if $G_{\mathbf{s}}(\theta) \neq 0$.

Orbits. The polynomial $R(x^2) \in \mathbb{Z}[x]$ is monic and even, and by Lemma 3.1(i) its roots are the $2r$ secular eigenvalues, which are distinct. Let f be a monic irreducible factor of $R(x^2)$ in $\mathbb{Q}[x]$; we call the set of its roots an *orbit*. We say that a switching $\mathbf{s}$ *fails at* an orbit O if $G_{\mathbf{s}}(\theta) = 0$ for some $\theta \in O$.

Lemma 3.3 (orbits). *Let f be as above, θ a root of f , and $t = \theta^2$.*

- (a) *If $f(-x) = f(x)$, then $\theta \notin \mathbb{Q}(t)$. We call the orbit of f even. (As 0 is not a root of $R(x^2)$, the case $f(-x) = -f(x)$ does not occur.)*
- (b) *If $f(-x) \neq f(x)$, then $\theta \in \mathbb{Q}(t)$, and $\pm f(-x)$ is another irreducible factor of $R(x^2)$, whose orbit is the negative of the orbit of f . We call these two orbits a non-even pair. If $\deg f = 1$, then $\theta \in \mathbb{Z}$ and t is a perfect square that is a root of R (an integer pair); otherwise $t \notin \mathbb{Q}$ (an irrational pair).*
- (c) *If an integer root t of R is not a perfect square, then $\{\sqrt{t}, -\sqrt{t}\}$ is an even orbit.*
- (d) *Let N_I and N_{II} be the numbers of integer and of irrational pairs, and N_{ei} the number of roots of R that are even integers but not perfect squares. Then $N_I + 2N_{II} + N_{ei} \leq r$.*
- (e) *For every switching $\mathbf{s}$, if $\mathbf{s}$ fails at an orbit O , then $G_{\mathbf{s}}$ vanishes on all of O .*

Proof. Let $h \in \mathbb{Q}[t]$ be the minimal polynomial of t . Then $h \mid R$ and $f \mid h(x^2)$. If $\theta \notin \mathbb{Q}(t)$, then $\deg f = [\mathbb{Q}(\theta) : \mathbb{Q}] = 2 \deg h$, so $f = h(x^2)$ is even. If $\theta \in \mathbb{Q}(t)$, then $\deg f = \deg h$; if f were even, say $f = f_1(x^2)$, then $f_1(t) = 0$ would give $\deg f = 2 \deg f_1 \geq 2 \deg h$, which is impossible. So f is not even, and $\pm f(-x)$, which divides $R(x^2)$ because $R(x^2)$ is even, is a different irreducible factor. If $\deg f = 1$, then θ is a rational root of the monic integer polynomial $R(x^2)$, hence an integer. If $\deg f \geq 2$, then $\deg h = \deg f \geq 2$ and $t \notin \mathbb{Q}$. This proves (a) and (b), and (c) holds because $x^2 - t$ is irreducible. For (d), the minimal polynomials of the roots of R are distinct irreducible factors of R : each integer pair uses a factor $t - q^2$, each irrational pair a factor h of degree at least 2 with $h(x^2) = \pm f(x)f(-x)$, and each root counted by N_{ei} a factor $t - m$ with m not a square. Distinct pairs give distinct factors, because a common root t' of the factors of two pairs would put $\pm\sqrt{t'}$ into both pairs, and $t - q^2 \neq t - m$ since m is not a square; so their degrees add up to at most $\deg R = r$. For (e), write $G_{\mathbf{s}}(x) = g(x) / \prod_{b \in B} (x^2 - b)$ with $g \in \mathbb{Q}[x]$; the denominator does not vanish at secular eigenvalues, and if $g(\theta) = 0$ for one root θ of f , then $f \mid g$. $\square$

The following reduction is the starting point of the proof.

Lemma 3.4 (equal signs next to the centre). *Suppose that $\sigma_1 = \dots = \sigma_k = \sigma$. Then $G_{\mathbf{s}}(\theta) = \sigma\theta + P_{\mathbf{s}}(t)$ with $P_{\mathbf{s}}(t) = \sum_{b \in B} A_b/(t - b) \in \mathbb{Q}(t)$ for every secular θ . Hence $\mathbf{s}$ fails at no even orbit, and $\mathbf{s}$ fails at a non-even pair, that is, at one of its two orbits, if and only if $P_{\mathbf{s}}(t) \in \{\theta, -\theta\}$, where θ is any root of the pair and $t = \theta^2$.*

Proof. Here $S_b = \sigma k_b$, so $\sum_b S_b \theta / (t - b) = \sigma \theta F(t) = \sigma \theta$. In an even orbit $\theta \notin \mathbb{Q}(t)$, so $\sigma \theta + P_{\mathbf{s}}(t) \neq 0$. The last statement holds because, by Lemma 3.3(e), $\mathbf{s}$ fails at the orbit of θ exactly when $G_{\mathbf{s}}(\theta) = 0$, which means $P_{\mathbf{s}}(t) = -\sigma \theta$, and at the orbit of $-\theta$ exactly when $G_{\mathbf{s}}(-\theta) = 0$, which means $P_{\mathbf{s}}(t) = \sigma \theta$. $\square$

4 Two rows of switchings

In this section we assume $b^* \geq 2$, that is, some branch has at least two leaves.

Standard leaf sums. For every branch i with $a_i \geq 1$ we fix a number λ_i° with $|\lambda_i^\circ| \leq a_i$ and $\lambda_i^\circ \equiv a_i \pmod{2}$, group by group, after ordering each I_b :

- if $b = 1$: $\lambda_i^\circ = -1$, except $\lambda_i^\circ = 1$ for the smallest $i \in I_1$ when $k_1 \geq 2$;
- if $b \geq 3$ is odd: $\lambda_i^\circ = -1$, except that the two smallest $i \in I_b$ get 1 and -3 when $k_b \geq 2$;

- if $b \geq 2$ is even: $\lambda_i^\circ = 0, -2, 0, -2, \dots$ along I_b .

Then (P1) for $b \geq 1$ with $k_b \geq 2$ the λ_i° with $i \in I_b$ are not all equal, and (P2) for $b \geq 2$ the smallest $i \in I_b$ has $|\lambda_i^\circ| < b$, so the leaves of that branch receive both signs whenever their sum is λ_i° . Put $\Lambda_b^\circ = \sum_{i \in I_b} \lambda_i^\circ$ for $b \geq 1$ and

$$U(t) = \sum_{b \in B, b \geq 1} \frac{\Lambda_b^\circ}{t - b}, \quad U_\beta(t) = U(t) - \frac{\Lambda_\beta^\circ}{t - \beta}.$$

Lemma 4.1 (realisable leaf sums). *Let $\beta \geq 1$ and $k \geq 1$, and let $\mathcal{L}_\beta(k)$ be the set of integers Λ for which there are $\lambda_1, \dots, \lambda_k \in \{-\beta, -\beta + 2, \dots, \beta\}$ with $\sum_i \lambda_i = \Lambda$, not all equal if $k \geq 2$, and, if $\beta \geq 2$, with $|\lambda_i| < \beta$ for some i . Then for $\beta \geq 2$, $\mathcal{L}_\beta(k)$ consists of the $\Lambda \equiv k\beta \pmod{2}$ with $|\Lambda| \leq k\beta - 2$, except $\Lambda = 0$ when $\beta = k = 2$; so $|\mathcal{L}_\beta(k)| = k\beta - 1 - [\beta = k = 2]$. For $\beta = 1$, $\mathcal{L}_1(k)$ consists of the $\Lambda \equiv k \pmod{2}$ with $|\Lambda| \leq k - 2$ if $k \geq 2$, and $\mathcal{L}_1(1) = \{\pm 1\}$; so $|\mathcal{L}_1(k)| = k - 1$ for $k \geq 2$ and $|\mathcal{L}_1(1)| = 2$.*

Proof. Every realisable Λ satisfies the stated parity and $|\Lambda| \leq k\beta$, and $|\Lambda| = k\beta$ would force all λ_i to equal $\pm\beta$. For $(\beta, k, \Lambda) = (2, 2, 0)$ the only candidates are $(0, 0)$ and $(2, -2)$, which are excluded. The case $\beta = 1$ is clear. Conversely, let $\beta \geq 2$ and let Λ be as stated. Put $M = (k\beta - \Lambda)/2$, the number of leaves of sign -1 ; then $1 \leq M \leq k\beta - 1$. For $k = 1$ take $\lambda_1 = \Lambda$. For $k \geq 2$ distribute M as evenly as possible, $m_i \in \{\lfloor M/k \rfloor, \lceil M/k \rceil\}$, and put $\lambda_i = \beta - 2m_i$. If $k \nmid M$, the m_i are not all equal and some m_i lies in $[1, \beta - 1]$. If $M = kq$, then $1 \leq q \leq \beta - 1$, and we use $(q + 1, q - 1, q, \dots, q)$ instead; some m_i lies in $[1, \beta - 1]$ unless $k = 2, q = 1$ and $\beta = 2$, which is the excluded case. $\square$

We write $\mathcal{L}_\beta = \mathcal{L}_\beta(k_\beta)$ for $\beta \in B$ with $\beta \geq 1$.

Lemma 4.2 (leaf row). *Let $\beta \in B$ with $\beta \geq 1$. For $\varepsilon \in \{\pm 1\}$ and $\Lambda \in \mathcal{L}_\beta$ let $\mathbf{s}_{\varepsilon, \Lambda}$ be a switching with $s_c = \varepsilon$, with $\sigma_i = 1$ for every branch (bare branches included), with leaf sums $\lambda_i = \lambda_i^\circ$ for $a_i \geq 1, i \notin I_\beta$, and with leaf sums on I_β as in Lemma 4.1 with total Λ (branch i gets $(a_i - \lambda_i)/2$ leaves of sign -1). These $2|\mathcal{L}_\beta|$ switchings form the leaf row of β . Every member satisfies (L) and (Z), and for every secular θ , with $t = \theta^2$,*

$$G_{\mathbf{s}_{\varepsilon, \Lambda}}(\theta) = \theta + \varepsilon + U_\beta(t) + \frac{\Lambda}{t - \beta}. \quad (4.1)$$

No member fails at an even orbit. An integer pair makes at most four members fail, at most two for each ε , and an irrational pair makes at most two members fail.

Proof. (L) holds by (P1) for $b \neq \beta$ and by Lemma 4.1 for $b = \beta$, since all $\sigma_i = 1$. For (Z): if $\beta \geq 2$, Lemma 4.1 gives a branch of size β with $|\lambda_i| < \beta$, whose leaves have both signs; if $\beta = 1$, the group $b^* \geq 2$ keeps its standard leaf sums, and (P2) gives such a branch. Since $A_b = \varepsilon k_b + \Lambda_b$ with $\Lambda_0 = 0$, Lemma 3.4 gives $G = \theta + \varepsilon F(t) + \sum_{b \geq 1} \Lambda_b/(t - b)$, which is (4.1), and no member fails at an even orbit. By Lemma 3.4, $\mathbf{s}_{\varepsilon, \Lambda}$ fails at a non-even pair if and only if $\varepsilon + U_\beta(t) + \Lambda w \in \{\theta, -\theta\}$, where $w = 1/(t - \beta) \neq 0$; for fixed ε the left-hand side is injective in Λ , so at most two Λ fail. If the pair is irrational, then $t \notin \mathbb{Q}$, so $w \notin \mathbb{Q}$ and $1, w$ are linearly independent over $\mathbb{Q}$. The member fails at the orbit of θ exactly when $\varepsilon + \Lambda w = -\theta - U_\beta(t)$, and at the orbit of $-\theta$ exactly when $\varepsilon + \Lambda w = \theta - U_\beta(t)$; each of these equations has at most one solution $(\varepsilon, \Lambda) \in \mathbb{Z}^2$. $\square$

Lemma 4.3 (bare row). *Suppose $k_0 \geq 1$. For $\varepsilon \in \{\pm 1\}$ and $0 \leq m \leq k_0$ let $\mathbf{s}_{\varepsilon, m}$ be a switching with $s_c = \varepsilon$, with leaf sums λ_i° on every branch with $a_i \geq 1$, with $\sigma_i = 1$ on these branches, and with $\sigma_i = -1$ on exactly m of the bare branches and $\sigma_i = 1$ on the others. These $2(k_0 + 1)$ switchings form the bare row. Every member satisfies (L) and (Z), and for every secular θ*

$$G_{\mathbf{s}_{\varepsilon, m}}(\theta) = \varepsilon + U(t) + \theta - \frac{2m}{\theta}. \quad (4.2)$$

An even orbit makes at most one member fail, and only if θ^2 is the same even integer for every θ in the orbit. An integer pair makes at most four members fail, and an irrational pair at most two.

Proof. (L) holds by (P1), and (Z) by (P2) applied to the group b^* . Now $S_0 = k_0 - 2m$ and $S_b = k_b$ for $b \geq 1$, so $\sum_b S_b/(t-b) = F(t) - 2m/t = 1 - 2m/t$, and (4.2) follows as before. In an even orbit, 1 and θ are linearly independent over $\mathbb{Q}(t)$, so $G = (\varepsilon + U(t)) + \theta(1 - 2m/t)$ vanishes only if $t = 2m$ and $\varepsilon = -U(t)$; this determines the member. Let $(\theta, -\theta)$ be a non-even pair. The member fails at the orbit of θ if and only if $\varepsilon - 2m/\theta = -\theta - U(t)$, and at the orbit of $-\theta$ if and only if $\varepsilon + 2m/\theta = \theta - U(t)$. For fixed ε each equation has at most one solution m , so at most four members fail. If the pair is irrational, then $1/\theta \notin \mathbb{Q}$, so each equation has at most one solution $(\varepsilon, m) \in \mathbb{Z}^2$. $\square$

Corollary 4.4. *Let $b^* \geq 2$. Then $T(\mathbf{a})$ has a good switching if*

- (a') $|\mathcal{L}_\beta| \geq 2N_I + N_{II} + 1$ for some $\beta \in B$ with $\beta \geq 1$, or
- (b') $k_0 \geq 1$ and $2(k_0 + 1) > 4N_I + 2N_{II} + N_{ei}$.

In case (a') some member of the leaf row of β is good; in case (b') some member of the bare row.

Proof. By Lemma 4.2, at most $4N_I + 2N_{II}$ of the $2|\mathcal{L}_\beta|$ members of the leaf row fail at some orbit, so under (a') some member fails at none. It satisfies (S) because every secular eigenvalue lies in an orbit, and (L) and (Z) by Lemma 4.2; so it is good by Lemma 3.2. Under (b') the same argument applies to the bare row, with at most $4N_I + 2N_{II} + N_{ei}$ failing members: by Lemma 4.3 an even orbit spoils a member only if θ^2 is the same even integer t for all θ in it; then the orbit is $\{\pm\sqrt{t}\}$ with t a root of R that is not a square (otherwise its factor would have degree 1 and would not be even, Lemma 3.3(b)), so there are at most N_{ei} such orbits. $\square$

5 Proof of Theorem 1

5.1 No branch with two leaves

Proposition 5.1. *Let $T = T(\mathbf{a})$ have $n \geq 3$ vertices and $b^* \leq 1$, so that $T = T(1^{k_1}, 0^{k_0})$: the centre has k_0 leaf neighbours and k_1 neighbours with one further leaf each. Then T has a good switching, namely:*

- (a) if $k_1 = 0$ (the star K_{1,k_0} , $k_0 \geq 2$): $s_c = 1$, and -1 on one leaf (on two leaves if $k_0 = 4$) and $+1$ on the other leaves;
- (b) if $k_1 = 1$ and $k_0 \geq 2$: $s_c = 1$; -1 on one leaf neighbour of c and $+1$ on the others; $+1$ on the other neighbour v of c and -1 on the leaf of v ;
- (c) if $k_1 \geq 2$ and $k_0 \geq 2$: $s_c = 1$; -1 on one leaf neighbour of c and $+1$ on the others; $+1$ on the k_1 other neighbours of c ; -1 on one of their leaves and $+1$ on the others;
- (d) if $k_0 = 1$: one of the two switchings $\mathbf{s}^\pm$ with $s_c = \pm 1$, all $\sigma_i = 1$ and leaf sums λ_i° (defined as in Section 4);
- (e) if $k_0 = 0$ (a spider with legs of length 2): the switching $\mathbf{s}^+$.

Proof. In all cases we use Lemma 3.2.

(a) Here $R(t) = t - k_0$, so $t = k_0$; with $A_0 = k_0$ and $S_0 = E$, where $E \in \{k_0 - 2, 0\}$ is the sum of the signs of the leaves, $G = (k_0 + E\theta)/t = 1 + E/\theta$. $G = 0$ would need $E^2 = k_0$, and $(k_0 - 2)^2 = k_0$ only for $k_0 \in \{1, 4\}$, while $E = 0$ for $k_0 = 4$. (L) is void, and (Z) holds because the leaf signs are not all equal.

(b) Here $R(t) = t^2 - (k_0 + 2)t + k_0$ and $G = 1 + (k_0 - 2)/\theta + (\theta - 1)/(t - 1)$. Substituting $k_0 = t(t - 2)/(t - 1)$, which is $R(t) = 0$, gives $G = (\theta^2 - 2)(\theta^2 + \theta - 1)/(\theta(\theta^2 - 1))$. Now $t = 2$ is not a root, as $R(2) = -k_0$. If $\theta^2 + \theta - 1 = 0$, then $t = \theta^2 = 1 - \theta$ satisfies $t^2 - 3t + 1 = 0$, and subtracting this from $R(t) = 0$ gives $(k_0 - 1)(1 - t) = 0$, which is impossible because $k_0 \geq 2$ and $t \neq 1$. (L) is void, and (Z) holds as in (a).

(c) Here $R(t) = h(t) := t^2 - (1 + k_0 + k_1)t + k_0$. With $A_0 = k_0$, $S_0 = k_0 - 2$, $A_1 = 2k_1 - 2$ and $S_1 = k_1$, and using $F(t) = 1$, we get $G = P + \theta Q$ with $P = (t + k_1 - 3)/(t - 1)$ and $Q = (t - 2)/t$. If G vanishes at θ or at $-\theta$, then $P^2 = tQ^2$, that is, $f(t) := t(t + k_1 - 3)^2 - (t - 2)^2(t - 1)^2 = 0$. Since $h(0) = k_0 > 0$, $h(1) = -k_1 < 0$, $h(k_0 + k_1) = -k_1 < 0$ and $h(k_0 + k_1 + 1) = k_0 > 0$, the roots of h lie in $(0, 1)$ and in $(k_0 + k_1, k_0 + k_1 + 1)$; so h has no integer root and is irreducible over $\mathbb{Q}$. Hence $h \mid f$, that is, the remainder $c_1t + c_0$ of f modulo h vanishes, where

$$\begin{aligned} c_1 &= -k_0^3 - 3k_0^2k_1 + 6k_0^2 - 3k_0k_1^2 + 12k_0k_1 - 13k_0 - k_1^3 + 7k_1^2 - 12k_1 + 8, \\ c_0 &= k_0^3 + 2k_0^2k_1 - 6k_0^2 + k_0k_1^2 - 7k_0k_1 + 13k_0 - 4. \end{aligned}$$

As polynomials in k_0 , both have constant leading coefficients ∓ 1 , so a common root forces their resultant to vanish:

$$\text{Res}_{k_0}(c_1, c_0) = (k_1 - 1)(k_1^6 - 11k_1^5 + 60k_1^4 - 148k_1^3 + 224k_1^2 - 192k_1 + 64) = 0.$$

For $k_1 \geq 2$ the first factor is nonzero, and the sextic has no integer root (an integer root would divide 64, and none of $\pm 1, \pm 2, \dots, \pm 64$ is a root). So (S) holds. (L) holds because the leaves of the k_1 branches do not all have the same sign, and (Z) holds as in (a).

(d) Here $R(t) = t^2 - (k_1 + 2)t + 1$ has discriminant $k_1(k_1 + 4)$, strictly between $(k_1 + 1)^2$ and $(k_1 + 2)^2$, so R is irreducible, $N_I = 0$, and there is at most one non-even pair. By Lemma 3.4, $s^\pm$ fails at no even orbit, and $P_{s^+}(t) = 1/t + 2[k_1 \geq 2]/(t - 1)$ and $P_{s^-}(t) = P_{s^+}(t) - 2F(t) = P_{s^+}(t) - 2$. If both failed at the pair, then $P_{s^+}(t)$ and $P_{s^+}(t) - 2$ would both lie in $\{\theta, -\theta\}$, so $2 \in \{\pm 2\theta\}$ and $t = 1$, which is not a root. (L) holds by (P1). The eigenvalue 0 does not occur, by Lemma 3.1(iii), since $k_0 = 1$ and all $a_i \leq 1$.

(e) Here $R(t) = t - k_1 - 1$ and $P_{s^+}(t) = 2[k_1 \geq 2]/(t - 1)$. For $k_1 = 1$, $G = \theta \neq 0$. For $k_1 \geq 2$, $G(\mu) = \mu + 2/(\mu^2 - 1)$ for each secular eigenvalue μ , and $G(\mu) = 0$ means $\mu^3 - \mu + 2 = 0$. This covers both $\mu = \theta$ and $\mu = -\theta$, and both are eigenvalues. The polynomial $x^3 - x + 2$ has no rational root, so it is irreducible, and its discriminant is $-104 < 0$, so it has only one real root. But every conjugate of an eigenvalue of a symmetric rational matrix is again an eigenvalue, hence real; so no eigenvalue satisfies $\mu^3 - \mu + 2 = 0$. (Directly: $\mu^2 = k_1 + 1$ would give $\mu k_1 = -2$, so $\mu^2 = 4/k_1^2 < k_1 + 1$.) (L) holds by (P1). For (Z), each leaf group is a single leaf, and $\sum_i \varepsilon_i = \Lambda_1^\circ \in \{-1, 2 - k_1\}$ differs from $s_c = 1$. $\square$

5.2 Counting non-even pairs

Lemma 5.2. *Let $b^* \geq 1$, let $g = |\{0, 1, \dots, b^*\} \setminus B|$, so that $r = b^* + 1 - g$, and let sq be the number of integers $q \geq 1$ with $q^2 \leq b^* - 1$ and $q^2 \notin B$. Then $\text{sq} \leq \lfloor \sqrt{b^* - 1} \rfloor$, $\text{sq} \leq g$, $N_I \leq \text{sq} + 1$ and $N_I + 2N_{II} + N_{\text{ei}} \leq r$.*

Proof. By Lemma 3.1(i), the roots $t_1, \dots, t_{r-1}$ lie in $(b_1, b^*) \subseteq [0, b^*)$, and no root of R lies in B . An integer pair has $t = q^2$ with $q \geq 1$ (Lemma 3.3(b)); if $t \neq t_r$, then $q^2 \leq b^* - 1$ and $q^2 \notin B$. Hence $N_I \leq \text{sq} + 1$. The squares counted by sq are distinct elements of $\{1, \dots, b^* - 1\} \setminus B$, so $\text{sq} \leq g$; the bound $\text{sq} \leq \lfloor \sqrt{b^* - 1} \rfloor$ is clear, and the last inequality is Lemma 3.3(d). $\square$

5.3 Large branches

Proposition 5.3. *If $b^* \geq 13$, then (a') holds with $\beta = b^*$. Hence some member of the leaf row of b^* is good.*

Proof. With the notation of Lemma 5.2,

$$2(2N_I + N_{II}) = 3N_I + (N_I + 2N_{II}) \leq 3(sq + 1) + (b^* + 1 - g) \leq 2sq + b^* + 4 \leq 2\lfloor\sqrt{b^* - 1}\rfloor + b^* + 4.$$

For $b^* \geq 13$ we have $2\lfloor\sqrt{b^* - 1}\rfloor + 7 \leq b^*$: if $b^* \leq 16$, then $\lfloor\sqrt{b^* - 1}\rfloor = 3$; if $b^* \geq 17$, then $q = \lfloor\sqrt{b^* - 1}\rfloor \geq 4$ satisfies $q^2 + 1 \leq b^*$ and $q^2 + 1 - (2q + 7) = (q - 1)^2 - 7 > 0$. So $2(2N_I + N_{II}) \leq 2b^* - 3$, and since the left-hand side is even, $2N_I + N_{II} \leq b^* - 2$. By Lemma 4.1, $|\mathcal{L}_{b^*}| = k_{b^*}b^* - 1 \geq b^* - 1 \geq 2N_I + N_{II} + 1$, which is (a'). The good member of the leaf row is given by Corollary 4.4, that is, by the counts of Lemma 4.2. $\square$

For $b^* \leq 12$ we use a finite search instead.

5.4 Branches with two to twelve leaves

For a set $B \subseteq \{0, 1, \dots, b^*\}$ containing b^* , with $r = |B|$ and $sq = sq(B)$ as in Lemma 5.2, put

$$\nu(B) = \min(sq + 1, r), \quad M(B) = 2\nu(B) + \lfloor(r - \nu(B))/2\rfloor.$$

Lemma 5.4. *Let $2 \leq b^* \leq 12$, and suppose that $T(\mathbf{a})$ satisfies neither (a'), for any β , nor (b'). Then $|\mathcal{L}_\beta| \leq M(B)$ for every $\beta \in B$ with $\beta \geq 1$, and $2(k_0 + 1) \leq 3\nu(B) + r$ if $k_0 \geq 1$. Let $\mathcal{R}$ be the set of multisets $\mathbf{a}$ with $2 \leq b^* \leq 12$ that satisfy these two conditions. Then $\mathcal{R}$ is finite; it has 13,376 elements, and each of them has $n \leq 124$.*

Proof. By Lemma 5.2, $N_I \leq \nu(B)$ and $N_{II} \leq \lfloor(r - N_I)/2\rfloor$. The function $x \mapsto 2x + \lfloor(r - x)/2\rfloor$ is increasing, so $2N_I + N_{II} \leq M(B)$, and failing (a') gives $|\mathcal{L}_\beta| \leq 2N_I + N_{II} \leq M(B)$. Failing (b') gives $2(k_0 + 1) \leq 4N_I + 2N_{II} + N_{ei} = 3N_I + (N_I + 2N_{II} + N_{ei}) \leq 3\nu(B) + r$. There are finitely many sets B , and for each of them the first condition bounds every multiplicity k_b with $b \geq 1$, because $|\mathcal{L}_b(k)| \geq k - 1$ by Lemma 4.1, and the second bounds k_0 . The count of $\mathcal{R}$ and the bound on n are obtained by an enumeration, which the Lean kernel checks (Section 7). $\square$

Proposition 5.5. *Every multiset in $\mathcal{R}$ except $(2, 2)$, $(2, 0, 0)$ and (3) satisfies (a') or (b').*

This is the computer-assisted step of the proof. The Lean kernel checks it (Section 7); for the programs of version 1 see Section 6.

Proposition 5.6. *The following switchings are good; the three trees have 7, 6 and 5 vertices.*

- (a) $T(2, 2)$: $s_c = \sigma_1 = \sigma_2 = 1$; the leaves of v_1 get $+1, +1$ and those of v_2 get $-1, -1$.
- (b) $T(2, 0, 0) = D(2, 2)$ (immediate from the definitions), the double star with two leaves at each of its two central vertices: $s_c = 1$; $+1$ on the two leaf neighbours of c ; -1 on the third neighbour v of c , and $+1, -1$ on the two leaves of v .
- (c) $T(3) = K_{1,4}$, rooted at a leaf c : $s_c = 1$; $+1$ on the centre v of the star; $+1, -1, -1$ on the other three leaves.

Proof. (a) $R(t) = t - 4$, so the secular eigenvalues are ± 2 , and $G = (2 + 2\theta)/2 = 1 + \theta \in \{3, -1\}$. For (L), at $\theta = \pm\sqrt{2}$ the numbers $1 + 2/\theta$ and $1 - 2/\theta$ differ. For (Z), $k_0 = 0$ and the leaf signs are constant on each branch, with $\sum_i \varepsilon_i = 0 \neq 1 = s_c$. (b) $R(t) = (t - 1)(t - 4)$, and $G = (2 + 2\theta)/t + (1 - \theta)/(t - 2)$ takes the values 4, -2 , 1, 1 at $\theta = 1, -1, 2, -2$. (L) is void, and (Z) holds because the leaves of v have both signs. (c) $R(t) = t - 4$ and $G = \theta/(t - 3) = \theta \neq 0$; (L) is void, and (Z) holds because the three leaves adjacent to v have both signs. $\square$

These three trees are also covered by known results: $K_{1,4}$ is a star, $T(2, 0, 0)$ a double star, and $T(2, 2)$ has seven vertices.

5.5 End of the proof

Proof of Theorem 1. The tree K_1 is trivial, since the switching (1) is good, and K_2 is excluded. Let T have $n \geq 3$ vertices; then $T = T(\mathbf{a})$ for some $\mathbf{a}$ (Section 3). If $b^* \leq 1$, Proposition 5.1 applies. If $b^* \geq 13$, Proposition 5.3 applies. If $2 \leq b^* \leq 12$ and (a') or (b') holds, Corollary 4.4 applies. Otherwise $\mathbf{a} \in \mathcal{R}$ by Lemma 5.4, so $\mathbf{a}$ is (2, 2), (2, 0, 0) or (3) by Proposition 5.5, and Proposition 5.6 applies. $\square$

6 Computations

In version 2 every result of this paper is proved in Lean, and the finite search of Proposition 5.5 and the enumeration of Lemma 5.4 are checked by the Lean kernel (Section 7). Version 1 relied on programs for these steps; they and their logs remain in version 1.7.0 of [4], directory `certificates/switching/`, as independent cross-checks, and version 2 does not rely on them. Those of the referee agent of version 1 (see the disclosure) begin with `rv_`. They are written in Python with SymPy [7], except `swtrees.c` and `swtrees_print.c` (in C). Proposition 5.5 was checked by three separately written programs: `final_region.py`, which enumerates $\mathcal{R}$ set by set as in Lemma 5.4; `final_region_indep.py`, which enumerates a larger set of multisets; and the referee's `rv_region.py`, which shares no code with the other two, enumerates $\mathcal{R}$ in another way and computes N_I , N_{II} and N_{ei} exactly. The results of all three agree with Proposition 5.5. Independently of Sections 3–5, the referee's `rv_certify.py` searched, for each multiset in $\mathcal{R}$, for a switching that is good by Lemma 2.2, and found one in every case. The switchings of Propositions 5.1 and 5.6 were checked for small parameters by `five_refined.py`, `rv_exceptional.py` and `rv_families.py`, and the lemmas of Sections 3 and 4 were tested on small trees (`test_refined_rows.py`, `test_bounds.py`, `rv_spectrum.py`, `rv_rows.py`, `rv_endtoend.py`).

The bounds of Lemma 4.2 are attained: in $T(15)$ the integer pair of $x \pm 4$ makes four members of the leaf row of $\beta = 15$ fail, and in $T(18, 0, 0)$ the irrational pair of $x^2 \pm 3x - 6$ makes two members of the leaf row of $\beta = 18$ fail, for every choice of the members of these rows. These two facts are proved in Lean.

7 Formal verification

All the results are formally verified in Lean 4 (version 4.34.1) with Mathlib [6] (version v4.34.1, commit d13f23b7). Version 1 of this paper was released with a partial formalisation, the modules `SwitchingThmH.lean`, `SwitchingRowCount.lean`, `SwitchingDiam4Cubic.lean` and `SwitchingWalkProfile.lean` of version 1.7.0 of [4], which proved some abstract steps of the proof and Remark 8.3. For version 2 every result of the paper was formalised: each of its mathematical claims is a statement of Table 1, a definition, a step of a written proof, or an observation that is immediate from the definitions, such as the identification of a small tree or of a known family in the notation $T(\mathbf{a})$.

Statements before formal proofs. In short: the statements were written from the released paper, reviewed independently and frozen by their hash before the formal proofs were accepted, and the proofs import the frozen file without modifying it. In detail, the statements were written after the written proofs: they formalise a paper that had already been released. On 27 September 2026, after version 1 had been released, an agent wrote a Lean file of definitions and statements, the *statement file*, which proves none of its statements. An independent agent reviewed it against the paper, checked its definitions with programs of its own and recommended freezing it as it was; its SHA-256 hash is 4fe68753c6151791fe601d4d8e5935ae3e150a21469bb4d9cf4053a706574303. Version 2 of the file keeps its 30 statements and all

its definitions unchanged and adds eight statements: the signature version of Theorem 1, the equivalence of diameter at most 4 and radius at most 2 for trees, that $-s$ is good when s is, the bound $t_r \leq b^* + k$, the criterion for a single secular eigenvalue stated after Lemma 3.2, the two sharpness examples of Section 6, and Remark 8.2. An independent agent reviewed version 2; after two additions that the review proposed (the criterion for a single eigenvalue, and the existence of the rows in the sharpness examples), which the same agent checked, it was frozen on 29 September 2026 at 17:01 UTC by recording its SHA-256 hash 69332eca957449e7613a6c95fa6bda906ecb2fbf688e0dc6ae4bba59fedec715. The proofs were written in separate files that import the statement file and never modify it. Work on them began on 29 September 2026 before version 2 was frozen: a module of auxiliary lemmas was compiled at 16:25 UTC, and nine agents were launched between 16:27 and 16:29 UTC; until the freeze they compiled against version 1, whose 30 statements version 2 keeps unchanged. Five of the eight new statements (`GoodNeg`, `DiamRadius`, `Lemma_U1`, `Theorem1_signed` and `Remark_8_2`) were proved, under other names, from the draft of version 2 before the freeze; their text did not change at the freeze. The three statements that changed at the freeze (`Lemma_3_2_pointwise`, `Sharp_T15` and `Sharp_T18`) were proved after it, and the theorems `check_X` of all eight new statements were written after it.

What is proved. The statement file states 38 propositions, listed in Table 1; the file of checks (`Research/Backfill/Paper8/Check.lean`) proves each of them, by a theorem `check_X : X` for every name X of the table. The statement `Theorem1_mathlib` is Theorem 1 written with Mathlib’s definitions only. In ASCII notation (`forall`, `exists`, $\rightarrow$, $\leq$, $\wedge$ and $\vee$ for $\wedge$ and $\vee$, `Real` for $\mathbb{R}$, `SimpleGraph.Iso` for an isomorphism of graphs, `Top` for the complete graph, `Membership.mem S x` for $x \in S$, `Ne a b` for $a \neq b$ and `Finset.univ.sum x` for $\sum_i x_i$) it reads

```
def Theorem1_mathlib : Prop :=
  forall (V : Type) [Fintype V] [DecidableEq V] (T : SimpleGraph V) [DecidableRel T.Adj],
    T.IsTree -> T.ediam <= 4 -> IsEmpty (SimpleGraph.Iso T (Top : SimpleGraph (Fin 2))) ->
      exists s : V -> Real, (forall v, s v = 1 \vee s v = -1) /\
        forall theta : Real,
          Module.End.HasEigenvalue
            (Matrix.toLin' (diagonal s * T.adjMatrix Real * diagonal s)) theta ->
            exists x, Membership.mem (Module.End.eigenspace
              (Matrix.toLin' (diagonal s * T.adjMatrix Real * diagonal s)) theta) x /\
              Ne (Finset.univ.sum x) 0
```

that is: for every tree T on a finite vertex type whose diameter (Mathlib’s extended diameter) is at most 4 and which is not isomorphic to K_2 , there is a vector s of signs such that every real eigenvalue of $D_s A(T) D_s$ has an eigenvector whose coordinates do not sum to 0.

How the proofs work. The formal proofs follow the written ones, with the following main differences. Lemma 2.2 is proved by decomposing s in an orthonormal basis of eigenvectors of A and interpolating; on the way the proofs show that a rational matrix has the same rank over $\mathbb{Q}$ and over $\mathbb{R}$, and that a matrix with full column rank modulo a prime has full column rank over $\mathbb{Q}$. The eigenvalues and eigenspaces of $T(\mathbf{a})$ (Lemma 3.1), the conditions of Lemma 3.2 and the criterion after it are read off the equations (3.1); the proof of Lemma 3.2 does not use Lemma 3.1. The roots of R are located by the signs of R at the points of B and the intermediate value theorem, and the estimate $R(t) > 0$ for $t > b^* + k$ gives the bound $t_r \leq b^* + k$. Lemma 3.3 uses minimal polynomials over $\mathbb{Q}$, the splitting of f into its even and odd parts, $f(x) = E(x^2) + xO(x^2)$ (in place of the degree of $\mathbb{Q}(\theta)$ over $\mathbb{Q}(\theta^2)$), and the fact that a rational root of a monic integer polynomial is an integer; for (d), a non-even pair $\{f, \pm f(-x)\}$ gives the irreducible factor h of R with $h(x^2) = \pm f(x)f(-x)$. In Lemma 4.1 the leaves of sign -1 are distributed by filling the branches in order rather than as evenly as possible. The proofs import the modules `SwitchingThmH`, `SwitchingRowCount` and `SwitchingWalkProfile` of version 1: the counts in Lemmas 4.2 and 4.3 use `sol_subsingleton`, `indep_of_irrational` and `affine_injective`, Proposition 5.3 uses `row_budget`, Lemma 2.2 uses `aeval_mulVec_of_eigen`, and Remark 8.3

In the paper	Lean statement
Theorem 1	<code>Theorem1</code> , <code>Theorem1_mathlib</code>
signature version (Section 1)	<code>Theorem1_signed</code>
diameter at most 4 iff radius at most 2 (Section 1)	<code>DiamRadius</code>
Lemma 2.1 and the remark after it; $-s$ is good if s is	<code>Lemma_2_1</code> , <code>Lemma_2_1_good</code> , <code>GoodNeg</code>
Lemma 2.2	<code>Lemma_2_2</code>
trees of diameter at most 4 are the $T(\mathbf{a})$; R is monic of degree r	<code>TreeStructure</code> , <code>Secular_mononic</code>
Lemma 3.1 (i)–(iv); $t_r \leq b^* + k$	<code>Lemma_3_1_i</code> ... <code>Lemma_3_1_iv</code> , <code>Lemma_U1</code>
Lemma 3.2 and the remark after it	<code>Lemma_3_2</code> , <code>Lemma_3_2_pointwise</code>
Lemma 3.3 (a)–(e)	<code>Lemma_3_3_a</code> ... <code>Lemma_3_3_e</code>
Lemma 3.4	<code>Lemma_3_4</code>
standard leaf sums, (P1), (P2)	<code>StdLeafSums</code>
Lemmas 4.1, 4.2, 4.3; Corollary 4.4	<code>Lemma_4_1</code> , <code>Lemma_4_2</code> , <code>Lemma_4_3</code> , <code>Cor_4_4</code>
Proposition 5.1	<code>Prop_5_1</code>
Lemma 5.2, Proposition 5.3	<code>Lemma_5_2</code> , <code>Prop_5_3</code>
Lemma 5.4, Proposition 5.5	<code>Lemma_5_4</code> , <code>Prop_5_5</code>
Proposition 5.6	<code>Prop_5_6</code>
the bounds of Lemma 4.2 are attained (Section 6)	<code>Sharp_T15</code> , <code>Sharp_T18</code>
Remarks 8.2 and 8.3	<code>Remark_8_2</code> , <code>Remark_8_3</code>

Table 1: The 38 statements of the statement file (version 2), in the namespace `BackfillPaper8.Challenge`.

is derived from `eigen_nonorth`. In Proposition 5.1(c), and in case (d) with $k_1 \geq 2$, the formal proofs square as in the written proof of (c) and reduce modulo R ; this gives a relation $c_1 t + c_0 = 0$ with integers c_1 and $c_0 \neq 0$, which contradicts the irrationality of t . So the resultant of (c) is not needed, and in case (d) the formal proof shows that s^+ itself is good (for $k_1 = 1$ a factorisation suffices). Here t is irrational because the discriminant of $R(t) = t^2 - mt + c$ lies strictly between $(m-2)^2$ and m^2 and differs from $(m-1)^2$ in parity. Case (e) uses the direct argument given in its proof. Steps of the written proofs that the formal proofs replace, such as the resultant and the sextic in Proposition 5.1(c) and the discriminant of the cubic in (e), are not formally verified; the results they prove are.

The finite parts of Lemma 5.4 and Proposition 5.5 are a program written in Lean. It lists the multisets of $\mathcal{R}$; this list is proved to contain every element of $\mathcal{R}$ exactly once, and it has 13,376 entries. For every entry a test, which is proved sound, checks $n \leq 124$ and $b^* + k \leq 48$, so that every root q^2 of R with q an integer has $q \leq 6$ by the bound $t_r \leq b^* + k$. It then checks (a') or (b') with N_I replaced by the number of integers $1 \leq q \leq 6$ with $R(q^2) = 0$ and with the inequality $N_I + 2N_{II} + N_{ei} \leq r$. Where this fails, the test uses a second criterion: if no integer $1 \leq q \leq 6$ satisfies $R(q^2) = 0$ (so $N_I = 0$) and $|R(0)|$ is not a square, then $2N_{II} < r$, since otherwise R would be the product of the minimal polynomials h of the roots t of the irrational pairs, and $|h(0)|$ is a square for each of them ($h(0) = \pm f(0)^2$ by $h(x^2) = \pm f(x)f(-x)$, and $f(0)$ is an integer by Gauss's lemma); the test then checks (a') with this bound. The Lean kernel evaluates the test on ten chunks of the list, and every entry passes one of the two criteria or is one of the three exceptions of Proposition 5.5.

Theorem 1 finds a vertex of eccentricity at most 2 as a vertex of minimum eccentricity, builds an isomorphism of the tree with some $T(\mathbf{a})$ and transfers good switchings along isomorphisms of graphs; the signature version writes every signature of a tree as $\sigma_{uv} = t_u t_v$, where t_u is the product of the signs on the path from a fixed root to u .

Checks. For every statement of Table 1, the recorded axiom audits (`#print axioms`) report only the axioms `propext`, `Classical.choice` and `Quot.sound`. A static scan found no `sorry`, `admit`, `native_decide`, `bv_decide`, `implemented_by`, `extern`, `unsafe`, and no axiom declaration, in the proof files. The Lean package of this paper, 80 modules (the statement file, its tests, the 69 proof modules, the 8 modules of further tests and the check file), together with the three modules of version 1 that it imports, was built module by module from an empty directory, reusing only the pinned toolchain and the compiled Mathlib and its dependencies, and every module was checked with `leanchecker`, all with exit code 0, twice (on 29 September 2026, between 18:33 and 19:05 UTC). For the release, the whole Lean project of the repository was rebuilt from a clean directory and all its modules were checked again with `leanchecker` (29 September 2026, 19:06–20:24 UTC). The statement file, its tests, the proofs, the further tests and the check file are in version 1.11.0 of the repository [4], directory `lean/Research/Backfill/Paper8/`; the receipt of the two builds is in `certificates/switching/lean-package-v2/`, and the logs of the rebuild of the whole project are in `logs/`.

Definitions the reader must trust. The statement `Theorem1_mathlib` uses only Mathlib’s definitions. Apart from Mathlib’s graphs, trees, diameters, radii, isomorphisms, adjacency matrices, matrices, eigenvalues and eigenspaces, orthogonal projections, polynomials, minimal polynomials, intermediate fields, ranks, primes and the fields $\mathbb{F}_p$, multisets, cardinalities, finite sets, integer square roots, parity and squares, the statements use the following definitions, each at most a few lines long: `IsEigenvalue` (an abbreviation for Mathlib’s `Module.End.HasEigenvalue of Matrix.toLin’ M`), `IsMainEigenvalue`, `AllEigenvaluesMain`, `IsSwitching`, `switchMatrix`, `ProjNe` and `IsGood` (main eigenvalues, switchings, $D_s A D_s$, $P_\theta s \neq 0$ and good switchings); `TV`, `parentB`, `treeT` with its decidable adjacency, and `adjT` (the tree $T(\mathbf{a})$ and its adjacency matrix); `branchMS`, `Bset`, `kb`, `bstar` and `Ib` (the multiset of branch sizes, B , k_b , b^* and I_b); `secular`, `IsSecular`, `secVec`, `sc`, `sig`, `lam`, `Sb`, `Lamb`, `Ab`, `Gs`, `Ps`, `condS`, `condL` and `condZ` (R , secular values, x^θ , the data of a switching, G_s , P_s and the conditions of Lemma 3.2); `R2`, `IsOrbitFactor`, `IsEvenPoly`, `mirror`, `nonEvenPairs`, `NI`, `NII`, `Nei` and `QAdj` ($R(x^2)$, orbits, even orbits, non-even pairs, N_I , N_{II} , N_{ei} and $\mathbb{Q}(t)$); `posIn`, `lamStd`, `LamStdB`, `Ufun`, `Ubeta` and `Lset` (standard leaf sums, U , U_β and $\mathcal{L}_\beta(k)$); `IsLeafRowMember`, `IsBareRowMember`, `FailsAtPair`, `FailsAtOrbit`, `CritAat`, `CritA` and `CritB` (the rows, failure, and the criteria (a’) and (b’)); `mkSw`, `gapCount`, `sqCountB`, `nuB`, `MB`, `InRegion` and `regionR` (switchings of Section 5, g , sq , $\nu(B)$, $M(B)$ and $\mathcal{R}$); and `sw56a`, `sw56b`, `sw56c` (the switchings of Proposition 5.6). Every one of them is checked by known-answer tests on small cases from the paper, including instances that must come out false. The tests are in two files: the tests of the statement file, and a second file written after the proofs, which covers the data of switchings, G_s and P_s , the conditions of Lemma 3.2, N_I , N_{II} , N_{ei} , $\mathbb{Q}(t)$, U , U_β , the rows, failure, the criteria, the region and the switchings of Proposition 5.6; to compute N_I and N_{II} it uses lemmas of the proofs. For instance, it checks that the three exceptions of Proposition 5.5 satisfy neither (a’) nor (b’).

Faithfulness and non-claims. All matrices are real (in Lemma 2.2, integer matrices and their ranks over $\mathbb{Q}$ and $\mathbb{F}_p$); eigenvalues are real eigenvalues, which suffices because the matrices are symmetric. “Multiplicity” and “simple” refer to the dimension of the eigenspace. A multiset $\mathbf{a}$ is represented by a tuple $(a_1, \dots, a_k)$, and the numerical data are functions of its multiset of entries; each set I_b is ordered like $\{1, \dots, k\}$. An orbit is a set of real roots of a monic irreducible factor of $R(x^2)$ over $\mathbb{Q}$, and a non-even pair is the set of two such factors f and $\pm f(-x)$. In Lemma 3.2 the common sign ε_i of the leaves of a branch is written as λ_i/a_i . “At most four members fail” means: for every choice of one member for each parameter, the set of parameters whose member fails has at most four elements; the sharpness statements say that this set has exactly four, respectively two, elements, for every choice, and that such a row exists; the other counts of Lemmas 4.2 and 4.3 are read in the same way. The signature version is stated for

every symmetric choice of signs ± 1 on the edges of T ; Remark 8.2 is stated for a matrix C with entries 0 and 1. The agreement of the Lean definitions with those of the paper was checked by the reviewing agents, not by a machine.

Changes in version 2. Apart from this section, Section 6, the disclosure and the dates, version 2 differs from version 1 as follows. Computational claims for which we have no formal proof were deleted: that the conclusion of Theorem 1 holds for all 63,242,254 trees with 3 to 24 vertices and, by direct certificates, for every tree of diameter at most 4 with at most 44 vertices; the counts of trees with non-even pairs; counts of good switchings of the three trees of Proposition 5.6; and, in Remark 8.1, the values of the largest $\tau(T)$ for $n \leq 24$ and the example of a tree with 13 vertices that needs three switched vertices. Section 6 now lists the programs of version 1, which remain cross-checks, and the two sharpness facts, which are proved in Lean. The bound $t_r \leq b^* + k$, which version 1 used without stating it, is now stated and proved (Section 3), and so is the criterion for a single secular eigenvalue after Lemma 3.2, which the abstract and Section 1 use. The abstract and Section 1 now speak of 13,376 multisets rather than trees (different multisets can give the same tree; for example $(3, 0, 0)$ and $(2, 0, 0, 0)$ both give the double star with two and three leaves), and Section 1 no longer says that the search leaves exactly three trees. In Section 1 the parenthesis “(all trees of diameter 3)” was removed, the known classes are given in the notation $T(\mathbf{a})$, the fact that a regular graph has only one main eigenvalue is attributed to [9], and “Integer pairs are rare” was replaced by the bound of Lemma 5.2. The statements on the spectrum of $D_s A D_s$ and on rational roots of R were moved into the proofs of Lemmas 2.1 and 3.3. Some sentences were removed: that a double star has two representations (Section 3), a remark on the labels printed by our programs (Section 4), a remark on the case $b^* = 12$ (Section 5), the last sentence of Remark 8.2, and the explanation of Stanić’s criterion in Remark 8.3. Section 1 also reports the searches for version 2. The theorems are unchanged; the written proofs are unchanged apart from the moves and deletions listed above and some steps added to the proofs of Lemmas 2.2, 3.3 and 3.4 and of Corollary 4.4, and Lemma 4.3 is worded more precisely.

8 Further remarks and questions

Remark 8.1 (how many vertices must be switched?). For a tree T with $n \geq 3$ vertices that has a good switching, let $\tau(T)$ be the least value of $\min(|U|, n - |U|)$ over the sets U of vertices for which the switching that is -1 exactly on U is good (as $-\mathbf{s}$ is good when $\mathbf{s}$ is, U and its complement give the same answer). We do not know whether τ is bounded. Version 1 of this paper reported values of the maximum of τ over the trees with $n \leq 24$ vertices, computed by programs; they are not formally verified, and we do not claim them here.

Remark 8.2 (pairing in bipartite graphs). Lemma 3.4 reflects a general fact. Let G be bipartite, with $A = \begin{pmatrix} 0 & C \\ C^\top & 0 \end{pmatrix}$, let $\theta \neq 0$ be an eigenvalue whose minimal polynomial is even, and let $\mathbf{s}$ be rational. Then $P_\theta \mathbf{s} = 0$ if and only if $P_{-\theta} \mathbf{s} = 0$. Indeed, $E_{\pm\theta}$ consists of the vectors $(u, \pm C^\top u / \theta)$ with $u \in \ker(CC^\top - \theta^2)$, a space with a basis over $\mathbb{Q}(\theta^2)$, and $\mathbf{s} \cdot (u, \pm C^\top u / \theta) = a \pm \theta b$ with $a, b \in \mathbb{Q}(\theta^2)$, while $\theta \notin \mathbb{Q}(\theta^2)$.

Remark 8.3 (regular graphs). We do not treat regular graphs, but record a sufficient condition that is formalised in `SwitchingWalkProfile.lean`. Let A be real symmetric of order $n \neq 2$ with constant row sums k , and suppose that $(A^j)_{vv} = \sum_u w_u (A^j)_{uu}$ for all $j \geq 0$, for an index v and weights $w_u > 0$. Then $\mathbf{s} = \mathbf{1} - 2e_v$ is good for A : if $f(A)\mathbf{s} = 0$ for a polynomial f , then $f(k)\mathbf{1} = 2Me_v$ for $M = f(A)$, summing coordinates gives $(n - 2)f(k) = 0$, so $Me_v = 0$, and $0 = (M^2)_{vv} = \sum_u w_u \|Me_u\|^2$ forces $M = 0$; now take $f = \mu_A / (x - \theta)$. We do not claim the case of walk-regular graphs ($w_u = 1/n$) as new; compare Stanić’s criterion [11, Corollary 4.6].

Questions. The problem of [2] remains open for trees in general, beginning with the trees of diameter 5 and 6, and so does the question whether τ in Remark 8.1 is bounded.

Thanks are due to Saieed Akbari, Hitesh Kumar, Bojan Mohar and Shivaramakrishna Pragada for posing the problem studied here, and to the authors of [1] for the conjecture.

Tool and computational resource disclosure

Version 1 of this paper was produced on 26 and 27 September 2026 (UTC) in a workflow directed by the author, using AI coding agents: Anthropic Claude Code, with the model Claude Opus 5.5. Claude Code agents selected the problem, found the theorem and its proof, wrote the programs and the Lean code of version 1, carried out the literature searches described in the introduction, and drafted the text. An adversarial checking agent (Claude Code, the same model), launched by the agent that found the proof, re-derived Lemmas 4.2, 4.3 and 5.4 and Proposition 5.3 and repeated the finite search; it found one sign error in a displayed equation, which did not affect the count and was corrected. Its programs were not kept. An independent referee agent (Claude Code, the same model), which had not produced the work, then refereed all proofs and the correspondence between the Lean statements of version 1 and the text, recomputed the finite search, the exceptional trees and further checks with its own programs (Section 6), and repeated the literature search; its five minor corrections were applied. A second independent agent (Claude Code, the same model) then reviewed the paper, re-derived its proofs and recomputed the computations of the proof with its own programs; its minor corrections were applied. The programs, logs and Lean code of version 1 are in version 1.7.0 of the repository [4] (directories `certificates/switching/`, `lean/Research/` and `logs/`).

Version 2 was produced from 27 to 29 September 2026. A Claude Code agent (model Claude Opus 5.5) wrote version 1 of the statement file of Section 7, and an independent Claude Code agent reviewed it and checked its definitions with its own programs. OpenAI Codex (desktop application; models `gpt-6-astra` and `gpt-6-sol`) checked version 1 of the paper independently, re-deriving its proofs and recomputing the finite region, the three exceptions and the sharpness examples of Section 6 with its own programs; it found no error and pointed out, among other things, the unstated bound $t_r \leq b^* + k$ and the claims that version 2 had to formalise or delete. Codex’s usage quota then ran out, and Claude Code carried out the rest: it wrote version 2 of the statement file, which an independent Claude Code agent reviewed; on 29 September 2026 nine Claude Code agents wrote the Lean proofs, working in parallel and coordinated by a Claude Code session that also wrote a module of auxiliary lemmas, assembled the package and ran the checks described in Section 7. An independent Claude Code agent reviewed the Lean package and checked the account of it in Section 7 against the code. Another independent Claude Code agent, using the model Claude Fable 5.1, refereed a draft of version 2: it checked every proof, compared the 38 Lean statements with the text, and recomputed the region, the exceptions, the switchings of Propositions 5.1 and 5.6 and the sharpness examples with its own programs; its corrections, which concern wording and steps missing from written proofs, were applied. On 29 September 2026, after Claude’s usage credits ran out, OpenAI Codex reviewed version 2 of the paper and statement file, examined the Lean proof routes and checked the recorded validation evidence against the sources. This was a source and evidence review; the Lean checks were not rerun by Codex. Throughout, “we” refers to this workflow. The author is not a professional mathematician. The author has read version 2 of the paper and, using a side-by-side table, compared the statements of the Lean theorems with the results stated here. No human mathematician has reviewed this paper. The correctness of the results rests on the Lean proofs of the statements in Table 1 and on the agreement of those statements with the paper, which was checked by the reviewing agents; the written proofs are given for the reader. AI systems are not authors of this paper; the author takes full responsibility for its content.

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